

DAVISON AAF, VA, USA

WMO: 724037

Lat: 38.717N

Lon: 77.183W

Elev: 73

StdP: 14.66

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93728

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.9	18.4	-2.7	4.8	19.1	2.4	6.2	24.3	26.6	32.5	24.5	32.1	5.9	310	0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	95.3	75.2	92.7	74.3	90.2	73.5	78.5	89.1	77.3	87.4	76.1	85.5	7.1	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.7	134.8	82.9	74.6	129.6	81.6	73.2	123.9	80.2	42.0	89.9	40.6	87.5	39.4	85.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.0	18.4	15.5	DB	5.5	98.5	6.3	3.5	0.9	101.0	-2.7	103.1	-6.3	105.0	-10.8	107.5
			WB	5.0	81.4	5.7	2.2	0.9	83.0	-2.5	84.3	-5.6	85.6	-9.8	87.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.2	35.1	38.9	45.9	56.9	65.3	73.9	78.3	76.3	69.9	58.5	46.8	39.9
	DBStd	16.89	10.11	9.90	9.52	8.23	7.61	5.66	4.71	4.88	6.75	7.84	8.08	8.91
	HDD50	1532	476	330	195	25	1	0	0	0	0	15	153	337
	HDD65	4207	927	731	596	268	94	6	0	0	28	232	545	780
	CDD50	4176	15	20	68	232	474	717	878	816	597	278	58	23
	CDD65	1377	0	0	4	26	103	273	413	351	175	31	0	1
	CDH74	13670	0	5	68	353	1082	2689	4328	3371	1505	254	10	5
	CDH80	5694	0	0	18	102	381	1089	2045	1457	542	60	0	0
(14) Wind	WSAvg	4.9	6.1	6.2	6.8	5.9	4.5	4.2	3.8	3.5	3.7	4.0	5.0	5.3
(15) Precipitation	PrecAvg	41.6	3.0	2.5	3.6	3.1	3.7	4.1	4.0	3.6	4.1	3.3	2.4	3.3
	PrecMax	50.5	6.2	6.8	8.6	5.2	8.5	11.0	7.3	7.3	12.4	7.3	5.0	6.5
	PrecMin	31.9	1.3	0.6	0.0	1.7	0.9	1.3	1.6	1.4	0.0	0.0	0.3	1.6
	PrecStd	4.9	1.2	1.3	1.8	1.1	1.9	2.2	1.5	1.4	2.7	1.8	1.5	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.7	73.0	81.5	88.2	93.1	96.8	100.0	98.7	95.1	86.9	75.1	70.2
		MCWB	61.2	59.9	61.3	66.2	72.0	75.7	75.9	74.5	71.6	71.9	61.4	60.4
	2%	DB	63.1	65.6	73.8	82.6	88.5	92.7	96.2	93.6	90.1	80.8	69.8	64.2
		MCWB	56.2	54.4	57.7	63.7	70.9	74.6	75.8	74.1	72.2	68.7	59.5	58.4
	5%	DB	56.9	60.1	68.3	77.6	84.2	89.7	93.0	90.7	86.3	76.2	65.2	59.5
		MCWB	49.9	50.3	55.1	61.5	69.6	73.8	75.5	73.7	71.6	65.8	56.1	53.5
	10%	DB	50.4	54.4	62.9	73.0	80.4	86.5	90.2	87.9	82.2	72.4	61.2	54.1
		MCWB	44.2	46.4	52.0	58.8	66.7	72.3	74.6	72.9	69.3	63.5	53.3	47.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.8	61.8	64.0	69.5	75.8	79.3	81.7	79.7	77.3	76.1	65.8	63.9
		MCDB	66.5	71.1	76.1	79.0	86.8	90.6	92.7	90.2	85.9	80.7	70.6	67.0
	2%	WB	57.5	57.1	60.7	67.1	73.5	77.2	79.2	77.7	75.4	71.6	62.1	59.8
		MCDB	63.1	63.9	70.7	77.9	84.3	88.1	90.2	87.8	83.4	77.8	67.2	64.1
	5%	WB	50.5	51.5	57.1	64.3	71.4	75.8	77.7	76.4	73.8	68.9	58.5	54.7
		MCDB	55.3	56.4	65.9	74.8	80.4	86.0	88.4	85.8	81.2	74.2	63.3	59.0
	10%	WB	44.7	47.5	52.9	61.5	69.4	74.4	76.4	75.1	72.2	65.3	55.1	49.0
		MCDB	49.6	54.3	61.2	70.9	76.9	83.7	86.1	83.6	79.5	70.4	59.9	53.2
(35) Mean Daily Temperature Range	5% DB	MDBR	17.4	19.1	20.8	23.6	22.1	21.2	20.7	20.6	20.7	22.0	20.8	17.1
		MCDBR	25.4	27.3	30.0	30.3	27.9	24.7	24.5	24.4	24.6	25.8	26.7	23.3
		MCWBR	19.0	18.2	17.1	14.5	13.0	10.2	8.8	8.4	10.0	14.4	16.8	17.9
	5% WB	MCDBR	22.6	23.7	26.2	25.5	23.2	21.3	20.4	19.9	19.5	21.1	20.6	21.0
		MCWBR	19.2	17.9	17.2	14.3	11.7	10.2	9.2	8.3	9.2	13.0	15.7	17.9
(40) Clear-Sky Solar Irradiance	taub		0.307	0.322	0.356	0.404	0.443	0.501	0.522	0.517	0.428	0.360	0.324	0.309
	taud		2.494	2.433	2.399	2.284	2.204	2.082	2.043	2.039	2.263	2.457	2.518	2.522
	Ebn at Noon		277	287	286	277	267	251	244	242	259	269	268	268
	Edh at Noon		25	30	35	41	45	51	53	52	39	29	24	23
(44) All-Sky Solar Radiation	RadAvg		669	934	1243	1562	1765	1946	1903	1700	1387	1046	761	589
	RadStd		57	103	89	116	193	147	98	88	161	116	64	56

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (74 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+236	+140

Nomenclature: See separate page